

Prof. Dr. Boas Pucker

Phylogenetic Analyses

Availability of slides

- All materials are freely available (CC BY) - after the lectures:
 - eCampus: WBIO-A-08
 - GitHub: <https://github.com/bpucker/teaching>
- Questions: Feel free to ask at any time
- Feedback, comments, or questions: [pucker\[a\]uni-bonn.de](mailto:pucker[a]uni-bonn.de)



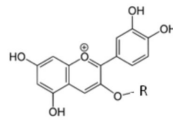
My figures and content can be re-used in accordance with CC BY 4.0, but this might not apply to all images/logos. Some figure were constructed using bioRender.com.

“Nothing in biology makes sense except in the light of evolution”

Phenylalanine



Anthocyanins



Anthocyanin



Antirrhinum majus

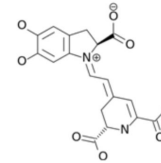
Tyrosine



Betalains



Mirabilis jalapa



Betalain

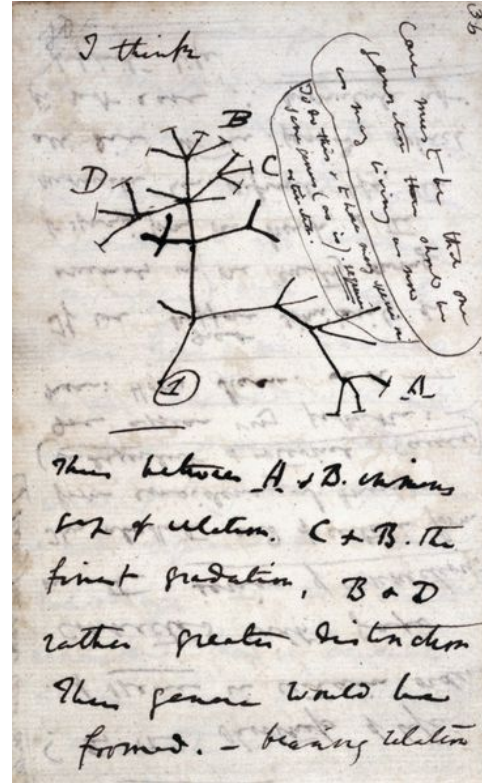
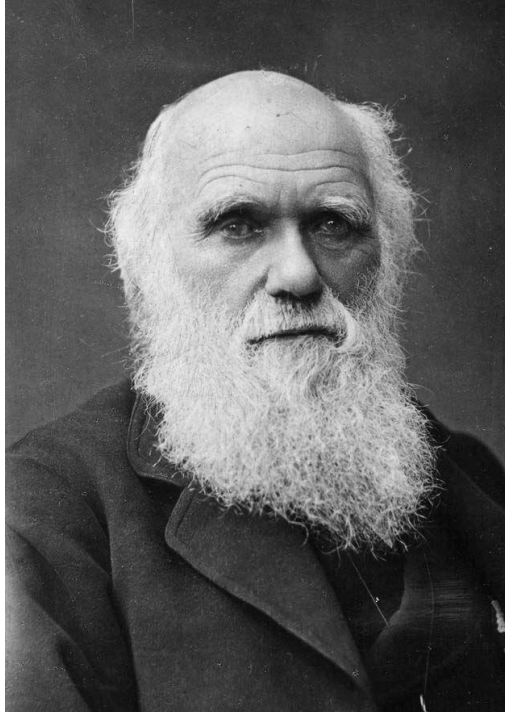


Anthocyanin color range



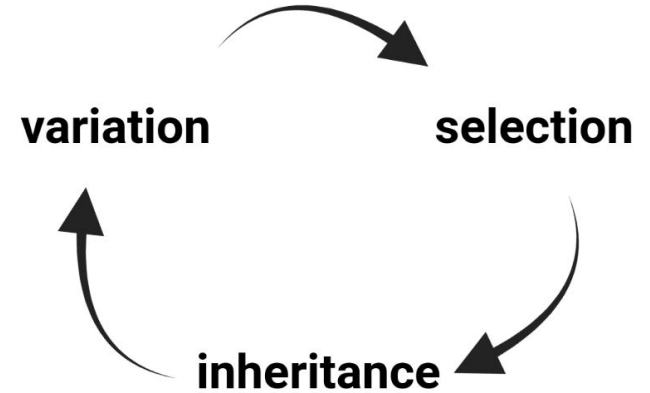
Betalain color range

Charles Darwin (1809-1882)

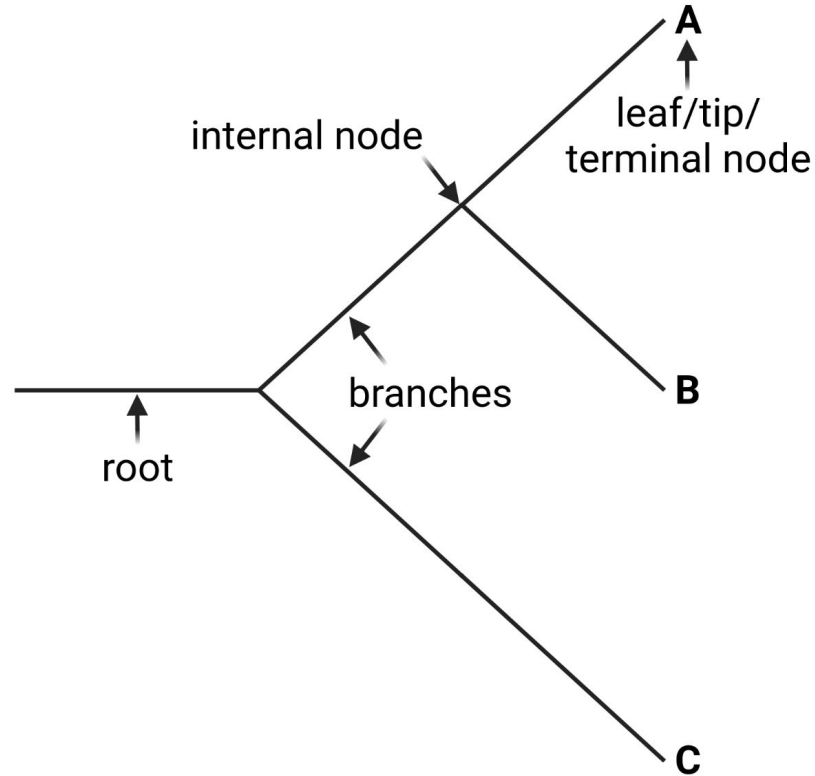


Basic evolution concept

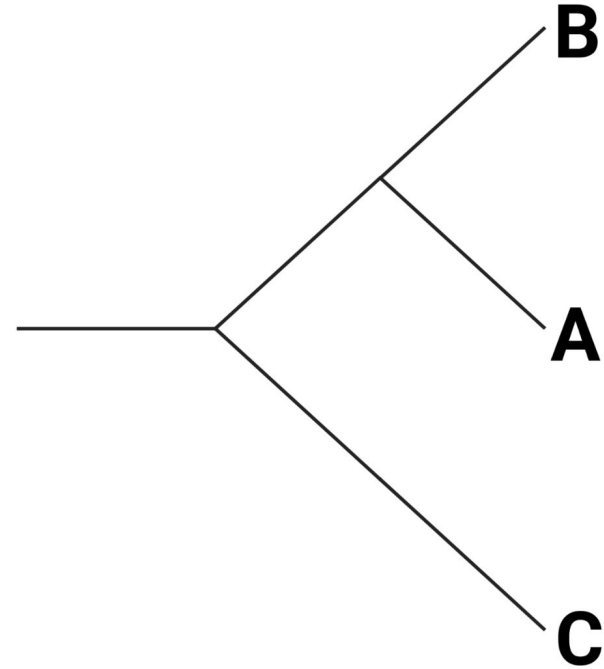
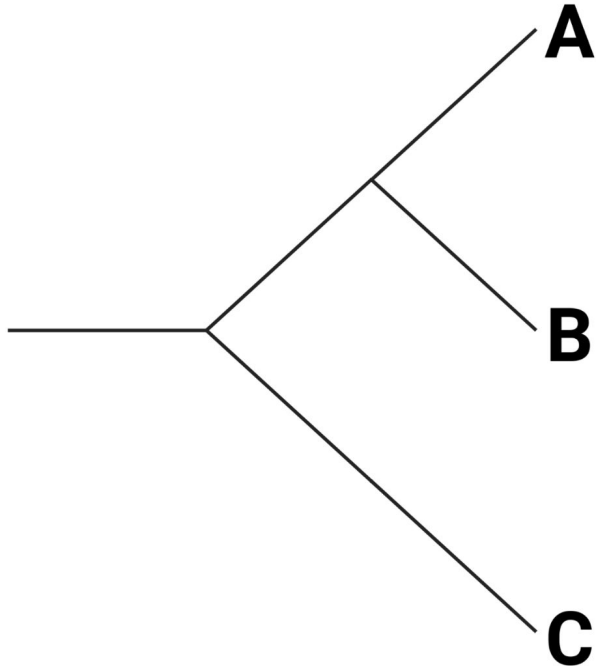
- Selection is performed by nature ('theory of natural selection')
- Selection requires (1) variation to act upon and (2) more offspring than sustainable
- Inheritance determines the properties of individuals (phenotype)



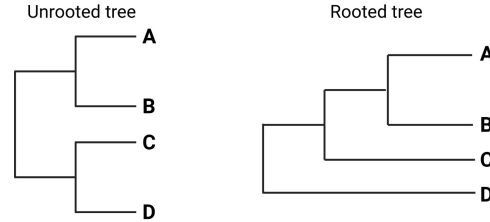
Elements of a phylogenetic tree



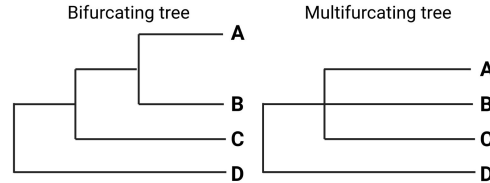
Tree topology and rotation



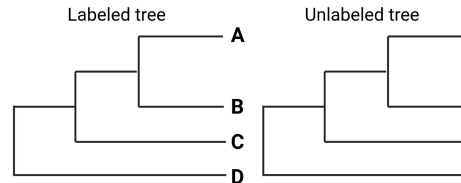
- Rooted vs. unrooted tree



- Bifurcating vs. multifurcating tree

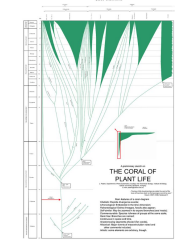
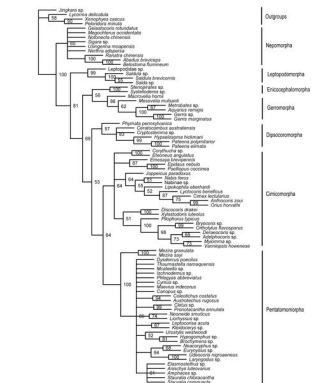
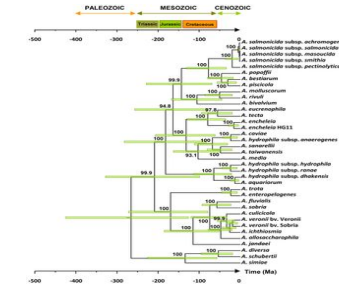
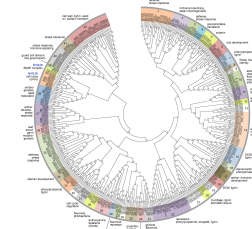


- Labeled vs. unlabeled trees



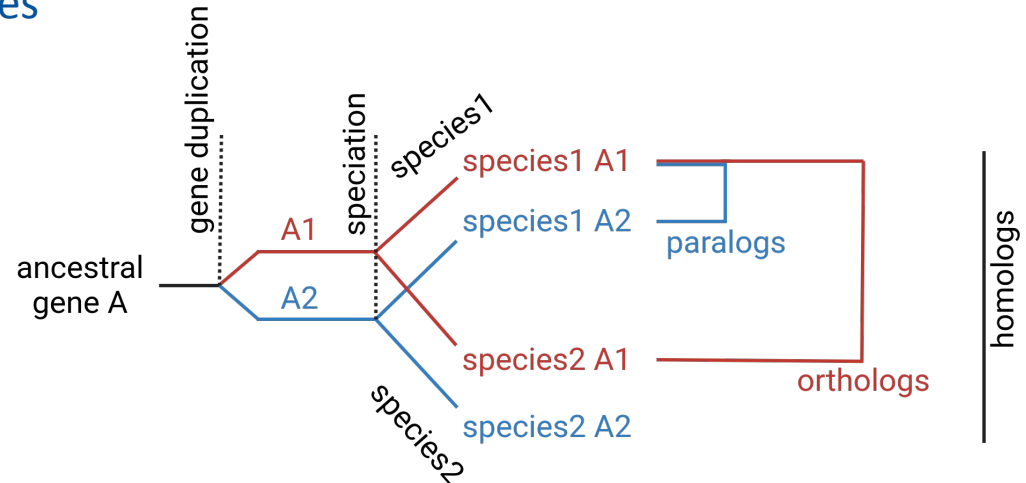
Types of special trees

- Cladogram: only represent the branching pattern without branch lengths
- Phylogram: branch lengths indicate number of changes
- Chronogram: branch lengths indicate time
- Spindle diagram: shows taxonomic diversity over time



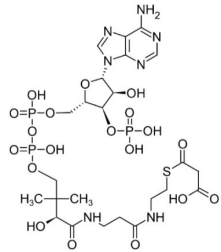
Homologs, orthologs, and paralogs

- Homolog = genes with common ancestors
- Ortholog = same gene in different species
- Paralog = gene copies in one species

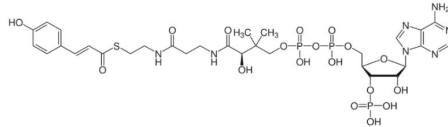


Example: orthologs

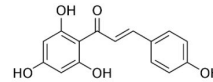
- CHS in different species has the same function
- Reaction is initial step of the flavonoid biosynthesis
- Flavonoid biosynthesis is conserved across plant species



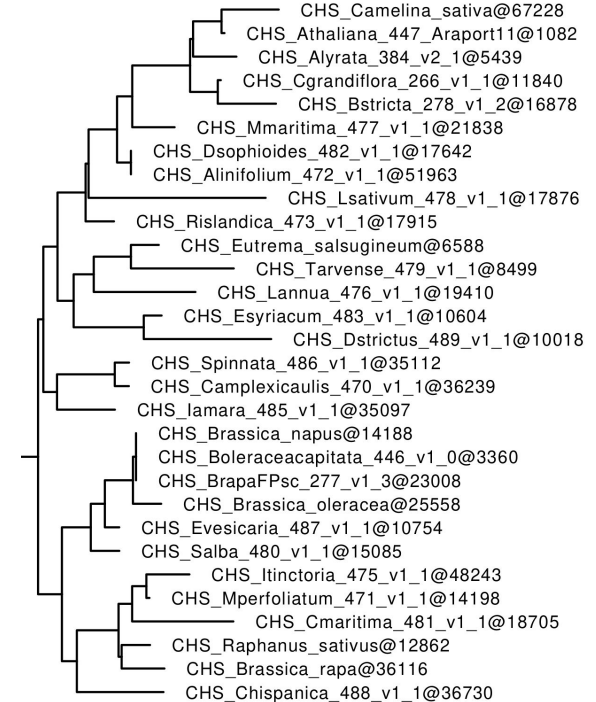
3x malonyl-CoA



4-coumaroyl-CoA

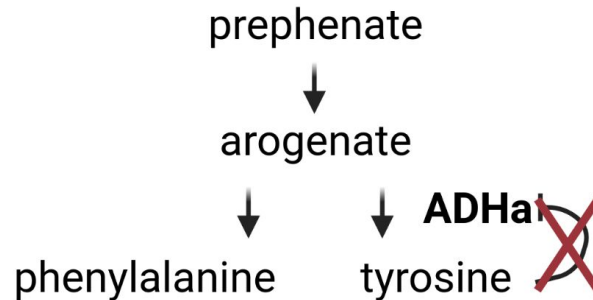
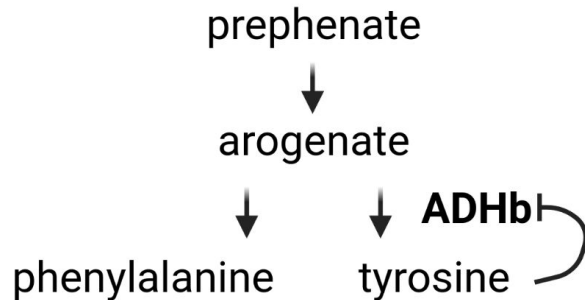


naringenin chalcone
(4x CoA + 3x CO₂)



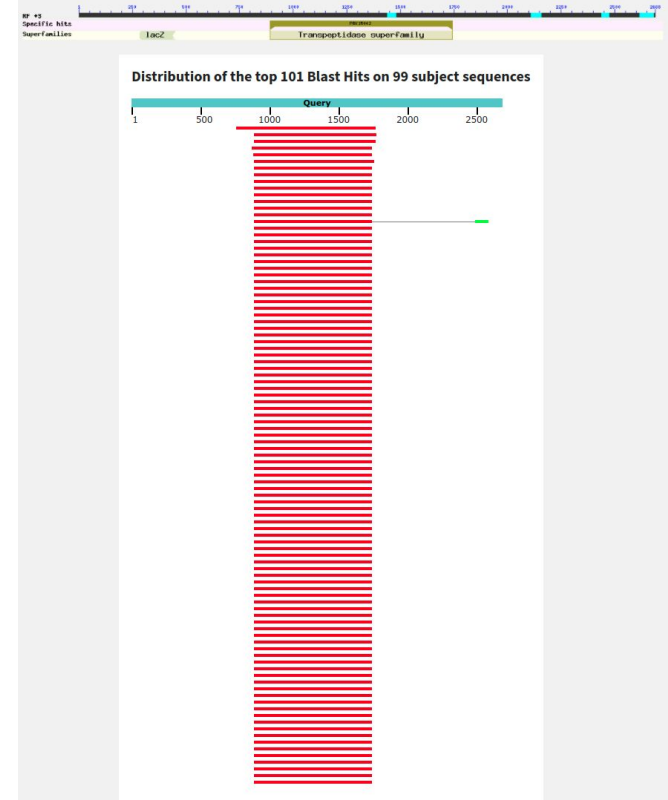
Example: paralogs

- ADH was duplicated in Caryophyllales
 - ADHa = feedback-resistance enzyme
 - ADHb = feedback-inhibited enzyme



BLAST - simple sequence searches

- BLAST = Basic Local Alignment Search Tool
- Identification of short stretches of sequence similarity
- Quick search against a large database
- Sequence similarity suggest homology i.e. common ancestors and similar function



NCBI website vs. local BLAST

Basic Local Alignment Search Tool

BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. [Learn more](#)

BLAST+ 2.13.0 is here!
Starting with this release, we are including the blastn_vdb and blastx_vdb executables in the BLAST+ distribution.
Thu, 17 Mar 2022 12:00:00 EST [More BLAST news...](#)

Web BLAST

Nucleotide BLAST
nucleotide → nucleotide

blastx
translated nucleotide → protein

tblastn
protein → translated nucleotide

Protein BLAST
protein → protein

BLAST Genomes

Enter organism common name, scientific name, or tax id

Human Mouse Rat Microbes

```

USAGE
blastn [-h] [-help] [-import_search_strategy filename]
[-export_search_strategy filename] [-task task_name] [-db database_name]
[-dbsize num_letters] [-glistlist filename] [-seqidlist filename]
[-negative_glistlist filename] [-negative_seqidlist filename]
[-taxids taxids] [-negative_taxids taxids] [-taxidlist filename]
[-negative_taxidlist filename] [-entrez_query entrez_query]
[-db_soft_mask filtering_algorithm] [-db_hard_mask filtering_algorithm]
[-subject subject_input_file] [-subject_loc_range] [-query input_file]
[-out output_file] [-evalue evalue] [-word_size int_value]
[-gapopen open_penalty] [-gapextend extend_penalty]
[-perc_identity float_value] [-qcov_hsp_perc float_value]
[-max_hsp int_value] [-xdrop_xmpar float_value] [-xdrop_gap float_value]
[-xdrop_gap_final float_value] [-searchsp int_value] [-penalty penalty]
[-reward reward] [-no_greedy] [-min_raw_gapped_score int_value]
[-template_type type] [-template_length int_value] [-dust_options]
[-filtering_db filtering_database]
[-window_masker taxid window_masker_taxid]
[-window_masker_db window_masker_db] [-soft_masking soft_masking]
[-ungapped] [-culling_limit int_value] [-best_hit_overhang float_value]
[-best_hit_score_edge float_value] [-subject_besbites]
[-window_size int_value] [-off_diagonal_range int_value]
[-use_index boolean] [-index_name string] [-icase_masking]
[-query_loc_range] [-strand strand] [-parse_deflines] [-outfmt format]
[-show_gis] [-num_descriptions int_value] [-num_alignments int_value]
[-line_length line_length] [-html] [-sorthits sort_hits]
[-sorthsp sort_hsp] [-max_target_seqs num_sequences]
[-num_threads int_value] [-remote] [-version]
  
```

DESCRIPTION
Nucleotide-Nucleotide BLAST 2.9.0v

NCBI website	Local BLAST
Convenient to use	Requires some command line knowledge
Requires no local computational resources	Computational resources needed
Transfer of query sequence	Query kept secret
Large databases available	Download of databases required

- HMMER: <http://hmmer.org/>
 - Dedicated to the identification of specific domains
- DIAMOND: <https://github.com/bbuchfink/diamond>
 - Much faster, but higher memory requirements

Local vs. global alignment

- Local alignment highlights small stretches with high similarity
 - random hit possible
 - better for strong sequence length differences
- Global alignment only works by overall high similarity
 - Full length similarity supports similar function

Local alignment



Global alignment



Multiple Sequence Alignment

MAFFT (Multiple Alignment using Fast Fourier Transform) is a high speed multiple sequence alignment program.

We have recently changed the default parameter settings for MAFFT. Alignments should run much more quickly and larger DNA alignments can be carried out by default. Please click the 'More options' button to review the defaults and change them if required.

Important note: This tool can align up to 500 sequences or a maximum file size of 1 MB.

STEP 1 - Enter your input sequences

Enter or paste a set of
AUTOMATIC

sequences in any supported format:

Or upload a file: [Browse...](#) No file selected.
Use a example sequence | Clear sequence | See more example inputs

STEP 2 - Set your Parameters

OUTPUT FORMAT
Pearson/FASTA

The default settings will fulfill the needs of most users.
[More options...](#) (Click here, if you want to view or change the default settings.)

STEP 3 - Submit your job

☐ Be notified by email (Tick this box if you want to be notified by email when the results are available)

Submit

```

-----
MAFFT v7.453 (2019/Nov/8)
https://mafft.cbrc.jp/alignment/software/
MBE 30:772-780 (2013), NAR 30:3059-3066 (2002)
-----

High speed:
% mafft in > out
% mafft --retree 1 in > out (fast)

High accuracy (for <~200 sequences x <~2,000 aa/nt):
% mafft --maxiterate 1000 --localpair in > out (% linsi in > out is also ok)
% mafft --maxiterate 1000 --genafpair in > out (% einsl in > out)
% mafft --maxiterate 1000 --globalpair in > out (% ginsi in > out)

If unsure which option to use:
% mafft --auto in > out

--op # :      Gap opening penalty, default: 1.53
--ep # :      Offset (works like gap extension penalty), default: 0.0
--maxiterate # : Maximum number of iterative refinement, default: 0
--clustalout : Output: clustal format, default: fasta
--reorder :   Outorder: aligned, default: input order
--quiet :     Do not report progress
--thread # :  Number of threads (if unsure, --thread -1)
--dash :      Add structural information (Rozewicki et al, submitted)

```

- Letters representing amino acid residues are arranged to highlight similarities
- Gaps (-) are inserted to show differences

Input FASTA file with sequences

```
>seq1
MDTEKYM EKWIDQGHALFP EEDQ
>seq2
MDTDIKYSEKWIQGHSLFPDDQ
>seq3
METEKYM EKWIQGH SIFPEDQ
>seq4
MDTEIKYM EKWIQGHALFPDDQP
```



Alignment of sequences

seq1	MDTE-KYM EKWIDQGHALFP EEDQ-
seq2	MDTDIKYSEKWI-QGHSLFPD-DQ-
seq3	METE-KYM EKWI-QGH SIFPE-DQ-
seq4	MDTEIKYM EKWI-QGHALFPD-DQP

CLUSTAL

```
AthCHS      MVMAGASSLDEIRQAQRADGPAGILAIGTANPENHVLQAEYPDYYFRITNSEHMTDLKEK
BvuCHS      M---ATPSVQEIRDAQRSNGPATILAIGTANPANEMYQAEYPDFYFRVTKSEHMSLKQK
            *  .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            *  .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:

AthCHS      FKRMCDKSTIRKRMHLTEEFLENPHMCAYMAPSLDTRQDIVVVEVPKLGKEAAVKAIK
BvuCHS      FKRMCDKSMIKKRYMHVTQELLEENPHMCDYNASSLNTRQDILATEVPKLGKEAAVKAIK
            ***** * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            ***** * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:

AthCHS      EWGQPKSKITHVVFCTTSGVDMPGADYQLTKLLGLRPSVKRLMMYQQGCFAGGTVLRIAK
BvuCHS      EWGQPRSKITHVIFCTTSGVDMPGADYQLTKLLGLRPSVKRFMLYQQGCFAGGTVLRIAK
            ***** .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            ***** .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:

AthCHS      DIAENNRGARVLVVCSEITAVTFRGPSDTHLDSLVGQALFSDGAAALIVGSDPDTSVGEK
BvuCHS      DIAENNRGARVLVVCSEITAVTFRGPSDTHLDSLVGQALFSDGAAALIVGSDPDTSVGEK
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:

AthCHS      PIFEMVSAQAQILPDSDGAIDGHLREVGLTFHLLKDVPLGKISKNIVKSLDEAFKPLGISD
BvuCHS      PIFQLVWAAQILPDSEGAIDGHLREVGLAFHLLKDVPLGKISKNIEKALVEAFKPIGIDD
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:

AthCHS      WNSLFWIAHPGGPAILDQVEIKLGLKEEKMRATHVLSEYGNMSSACVLFILDEMRRKSA
BvuCHS      WNSIFWAAHPGGPAILDQVESKLGKQDKLSTTRHVLSEFGNMSSACVLFILDEMRRKSM
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:

AthCHS      KDGVATTGEGLEWGVLFGFGPGLTVETVVLHSVPL--
BvuCHS      KEGMATTGEGLEWGVLFGFGPGLTVETVVLHSVPIAN
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
            * .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .: .:
```

FASTA

```
>AthCHS
MVMAGASSLDEIRQAQRADGPAGILAIGTANPENHVLQAEYPDYYFRITNSEHMTDLKEK
FKRMCDKSTIRKRMHLTEEFLENPHMCAYMAPSLDTRQDIVVVEVPKLGKEAAVKAIK
EWGQPKSKITHVVFCTTSGVDMPGADYQLTKLLGLRPSVKRLMMYQQGCFAGGTVLRIAK
DLAENNRGARVLVVCSEITAVTFRGPSDTHLDSLVGQALFSDGAAALIVGSDPDTSVGEK
PIFEMVSAQAQILPDSDGAIDGHLREVGLTFHLLKDVPLGKISKNIVKSLDEAFKPLGISD
WNSLFWIAHPGGPAILDQVEIKLGLKEEKMRATHVLSEYGNMSSACVLFILDEMRRKSA
KDGVATTGEGLEWGVLFGFGPGLTVETVVLHSVPL--

>BvuCHS
M---ATPSVQEIRDAQRSNGPATILAIGTANPANEMYQAEYPDFYFRVTKSEHMSLKQK
FKRMCDKSMIKKRYMHVTQELLEENPHMCDYNASSLNTRQDILATEVPKLGKEAAVKAIK
EWGQPRSKITHVIFCTTSGVDMPGADYQLTKLLGLRPSVKRFMLYQQGCFAGGTVLRIAK
DLAENNRGARVLVVCSEITAVTFRGPSDTHLDSLVGQALFSDGAAALIVGSDPDTSVGEK
PIFQLVWAAQILPDSEGAIDGHLREVGLAFHLLKDVPLGKISKNIEKALVEAFKPIGIDD
WNSIFWAAHPGGPAILDQVESKLGKQDKLSTTRHVLSEFGNMSSACVLFILDEMRRKSM
KEGMATTGEGLEWGVLFGFGPGLTVETVVLHSVPIAN
```

- Removal of alignment columns with low abundance
- Columns with low abundance are not informative
- Reduced alignment lengths makes tree construction easier

Alignment of sequences

seq1	MDTE-KYMEKWI	D	Q	G	H	A	L	F	P	E	D	Q	-
seq2	MDTDIKYSEKWI	-	Q	G	H	S	L	F	P	D	-	D	Q
seq3	METE-KYMEKWI	-	Q	G	H	S	I	F	P	E	-	D	Q
seq4	MDTEIKYMEKWI	-	Q	G	H	A	L	F	P	D	-	D	Q



Trimmed alignment of sequences

seq1	MDTE-KYMEKWI	Q	G	H	A	L	F	P	E	D	Q
seq2	MDTDIKYSEKWI	Q	G	H	S	L	F	P	D	D	Q
seq3	METE-KYMEKWI	Q	G	H	S	I	F	P	E	D	Q
seq4	MDTEIKYMEKWI	Q	G	H	A	L	F	P	D	D	Q

Construction of a phylogenetic tree

- Construction of phylogenetic trees is based on sequence alignments
- Similar sequences (recent shared ancestors) are grouped

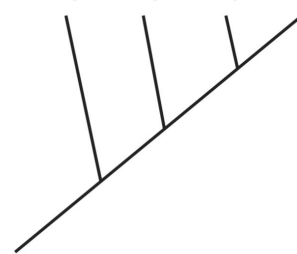
Trimmed alignment of sequences

seq1	MDTE-KYMEKWIQGHALFPEDQ
seq2	MDTDIKYSEKWIQGHSLFPDDQ
seq3	METE-KYMEKWIQGHSIFPEDQ
seq4	MDTEIKYMEKWIQGHALFPDDQ

Distance calculation

	seq1	seq2	seq3	seq4
seq1	.	5	3	2
seq2	.	.	6	3
seq3	.	.	.	5
seq4	.	.	.	.

seq2 seq3 seq1 seq4



Substituting amino acids with codons

- Phylogenetic signal is boosted by substitution of amino acids by codons
- More positions that can show variation
- Alignment can still be based on amino acid residues (avoids splitting codons)

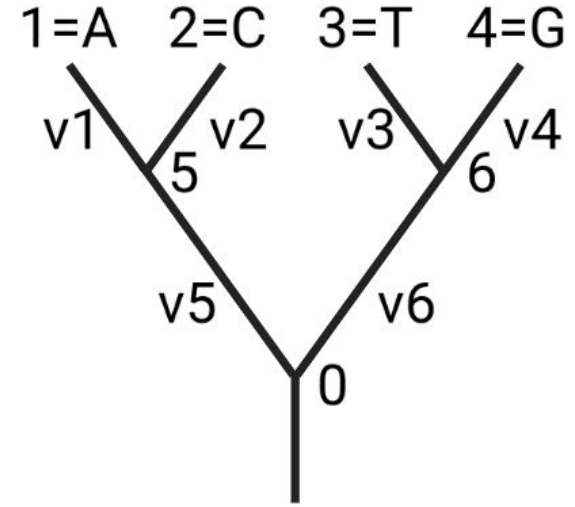
ATG GAT ACT GAA ...



seq1	MDTE-KYMEKWIDQGHALFPPEEDQ-
seq2	MDTDIKYSEKWI-QGHSLFDPD-DQ-
seq3	METE-KYMEKWI-QGHSIFPE-DQ-
seq4	MDTEIKYMEKWI-QGHALFPD-DQP

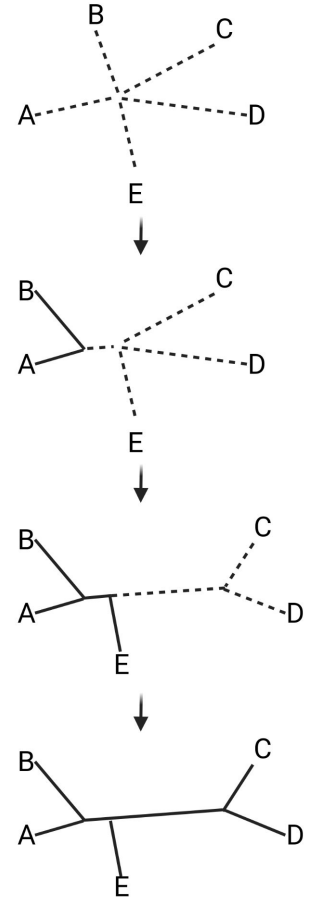
Maximum Likelihood (ML) trees

- Finding the tree with highest likelihood to explain the observed sequences
- Number of possible trees depends on number of sequences
- Often impossible to explore all possible trees (very computationally demanding)
- Tool: RAxML-NG



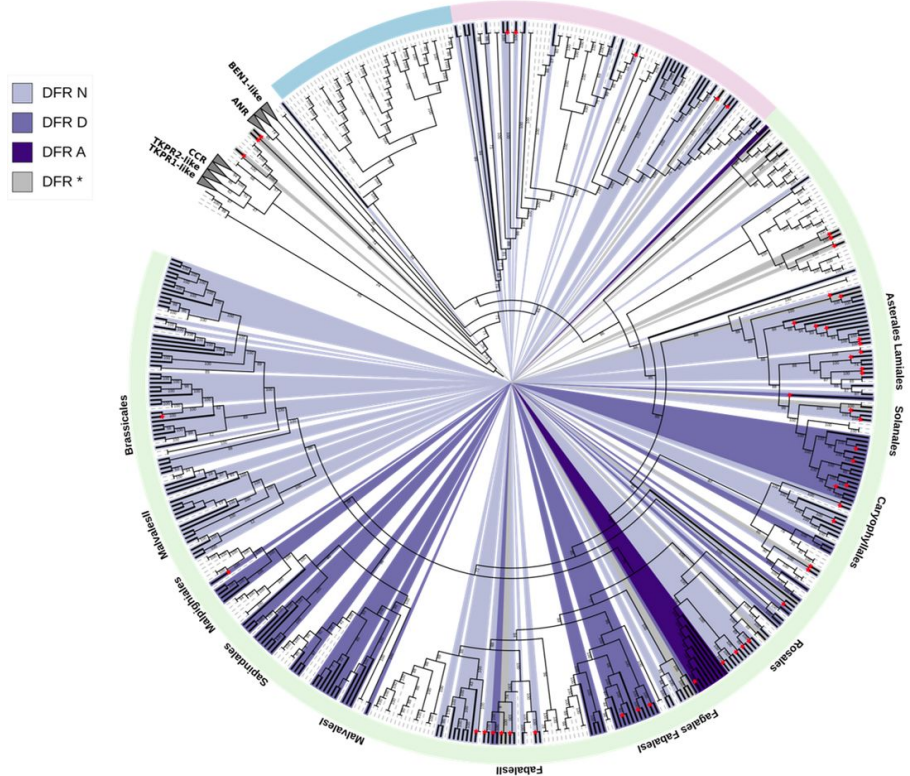
Neighbor Joining (NJ) tree

- Combination of sequences (groups of sequences) with smallest distance
- Quick and computationally cheap approach
- Less reliable than maximum likelihood (ML) trees
- Tool: MEGA



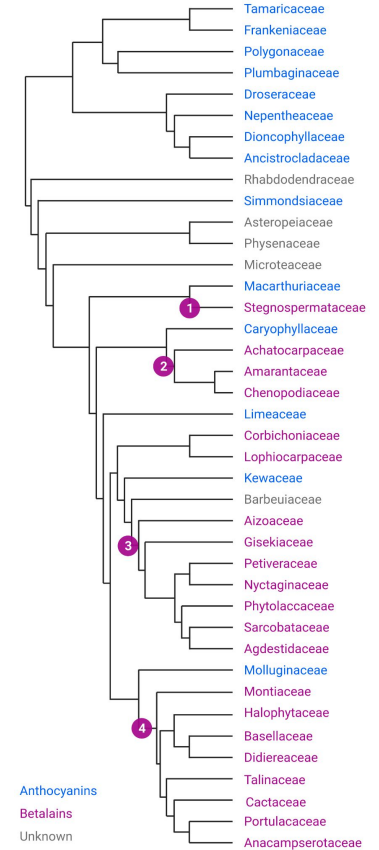
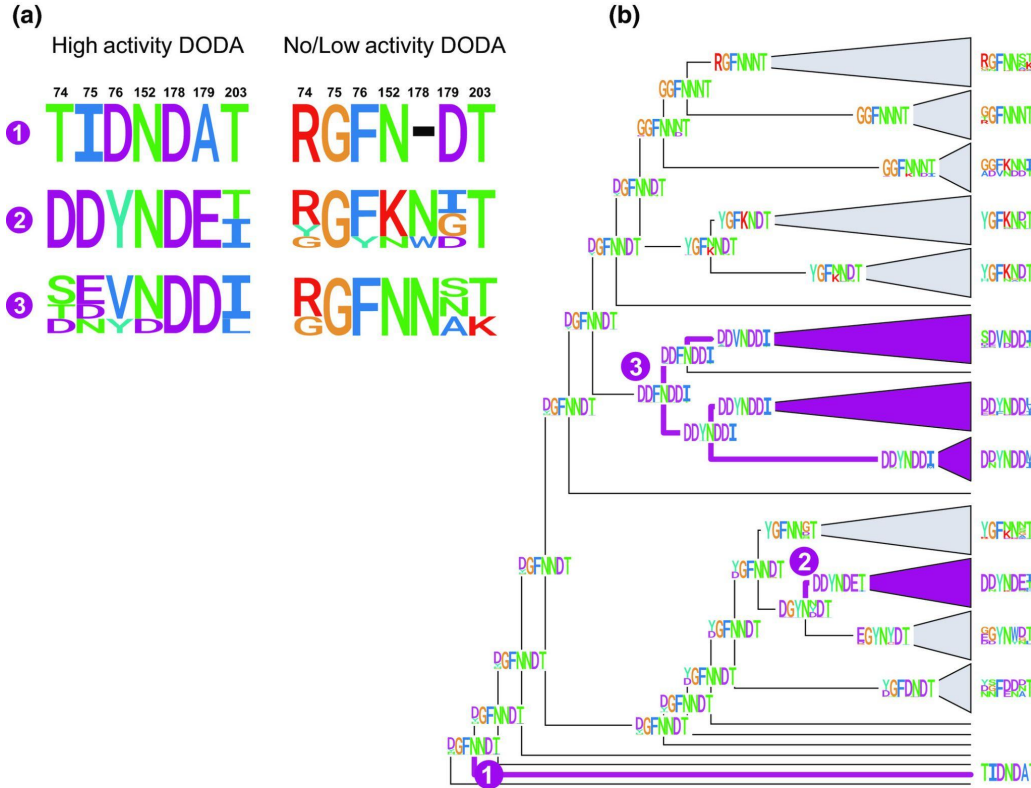
EXAMPLE: gene tree of *DFR*

- Evolution of *DFR* in land plants
- Multiple clades visible
- Sequences of closely related species cluster in lineages

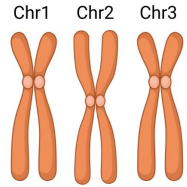


-

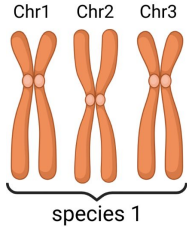
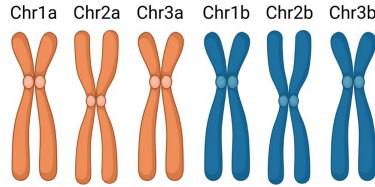
Example: Ancestral state reconstruction



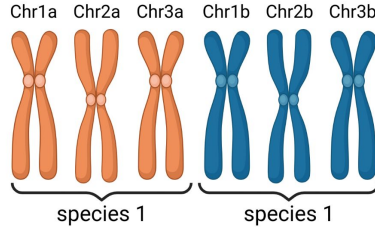
Whole genome duplications



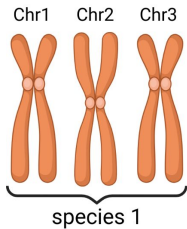
whole
genome
duplication



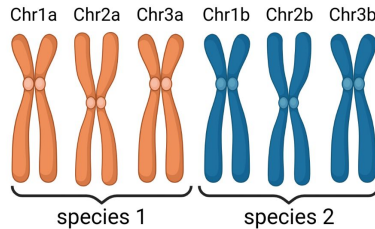
whole
genome
duplication



autopolyploidy



whole
genome
duplication

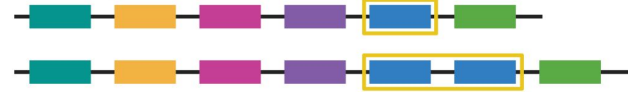


allopolyploidy

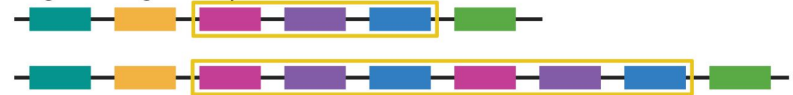
Classification of duplications: tandem, segment, WGD

- Gene copies can be the results of different mechanisms:
 - Tandem duplication: copies are located next to each other on the same chromosome
 - Segment duplication: like tandem duplications, but multiple adjacent genes are also duplicated
 - Whole genome duplication (WGD): all genes are duplicated

Tandem gene duplication



Segmental gene duplication



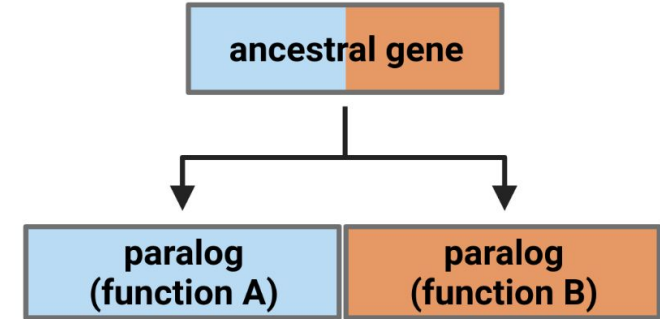
Whole genome duplication



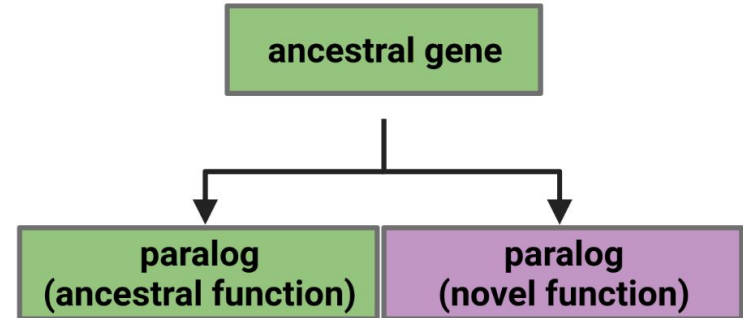
Subfunctionalization & Neofunctionalization

- Subfunctionalization: gene copies fulfill parts of the ancestral function
- Neofunctionalization: one copy maintains ancestral function, while other copy can acquire a novel function
- Observation bias: gene loss following gene duplications is invisible and probably most likely event

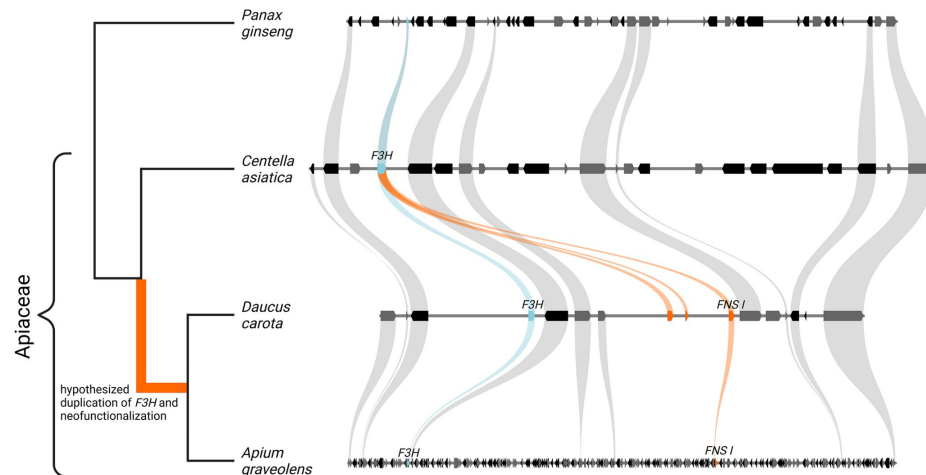
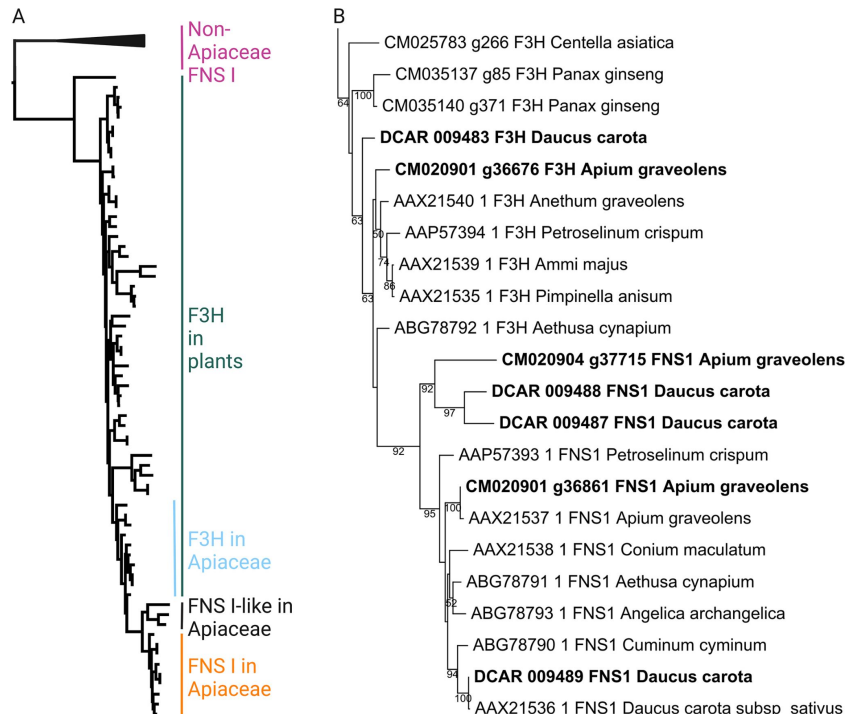
Subfunctionalization



Neofunctionalization



EXAMPLE: *F3H/FNS I* / neofunctionalization



- Concepts of evolution
- Homology (orthology & paralogy)
- Alignment and phylogenetic tree construction
- Ancestral state reconstruction
- Neo- and subfunctionalization

Time for questions!

1. What is the difference between homolog, paralog, and ortholog?
2. What are the advantages/disadvantages of running BLAST locally vs. at the NCBI website?
3. What are suitable BLAST alternatives?
4. What is the difference between a local and a global alignment?
5. What are the different methods for the construction of a phylogenetic tree?
6. What is ancestral state reconstruction?
7. What are possible mechanisms underlying gene duplications?
8. What is neofunctionalization/subfunctionalization?
9. What is the relationship of A and B? (paralog/ortholog)
10. What is the relationship of D and E (paralog/ortholog)

